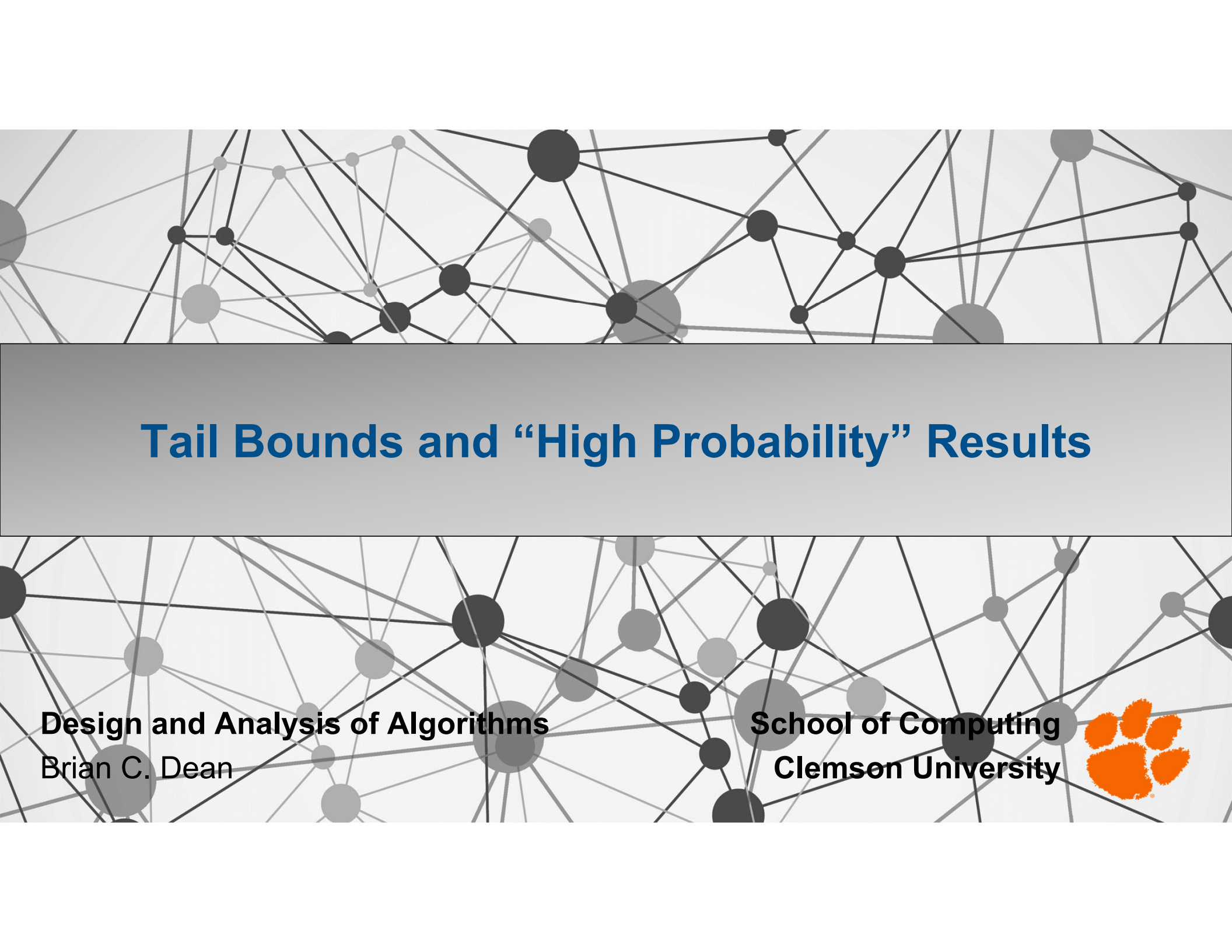


The background of the slide features a complex network diagram. It consists of numerous circular nodes of varying sizes, some in dark gray and others in light gray, interconnected by a web of thin, dark gray lines. The nodes are distributed across the entire slide, with a higher density in the upper and lower portions, framing the central title area.

Tail Bounds and “High Probability” Results

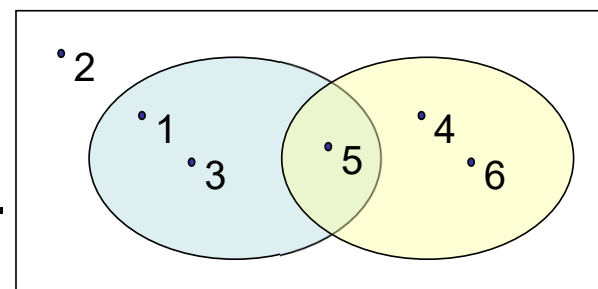
Design and Analysis of Algorithms
Brian C. Dean

School of Computing
Clemson University



Intersections and Unions of Events

- If A and B are events (sets of outcomes), then
 - $A \cap B$ is the event that both occur
 - $A \cup B$ is the event that either A or B occur, or both.
- **$\Pr[A \cup B] = \Pr[A] + \Pr[B] - \Pr[A \cap B] \leq \Pr[A] + \Pr[B]$** , as is clear from the **Venn diagram** below.
 - Related: the principle of inclusion-exclusion
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- A and B **independent** iff **$\Pr[A \cap B] = \Pr[A] \Pr[B]$** .
 - Application: boosting success probability with multiple independent runs

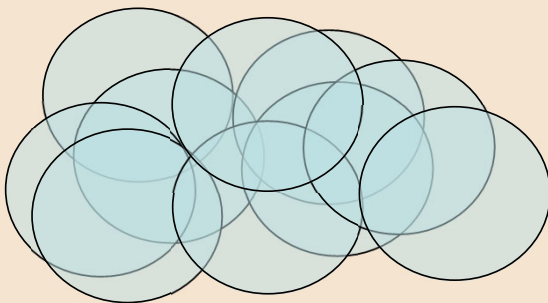


A: Roll an odd number
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$$\Pr[E_1 \cup E_2 \cup \dots \cup E_N] = \Pr[E_1] + \Pr[E_2] + \dots + \Pr[E_N]$$

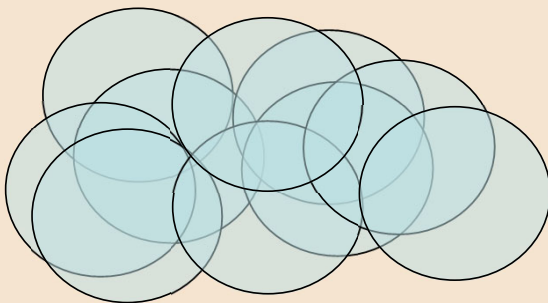


- Probabilities of all pairwise event intersections
- + Probabilities of all 3-tuples of event intersections
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$$\begin{aligned}\Pr[\text{entire machine fails}] &= \Pr[\text{part 1 fails} \cup \text{part 2 fails} \cup \dots \cup \text{part } N \text{ fails}] \\ &\leq \Pr[\text{part 1 fails}] + \Pr[\text{part 2 fails}] + \dots + \Pr[\text{part } N \text{ fails}] \\ &= 1/2\end{aligned}$$

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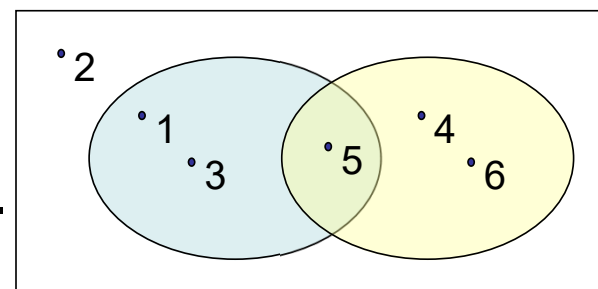
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Input consisting of N **elements**. $\Pr[\text{any one element "fails"}] \leq 1 / (2N)$.

$$\begin{aligned}\Pr[\text{entire algorithm fails}] &= \Pr[\text{element 1 fails} \cup \dots \cup \text{element } N \text{ fails}] \\ &\leq \Pr[\text{element 1 fails}] + \dots + \Pr[\text{element } N \text{ fails}] \\ &= 1/2\end{aligned}$$

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Example: High Probability Bounds

- Method used to process a file fails w/ probability $\leq 1/2$.
- Make $\log_2 N$ independent attempts to process one file:
 $\Pr[\text{failure on ALL attempts}] \leq (1/2)^{\log_2 N} = 1 / N$.
- We therefore succeed on this file with high probability.

Example: High Probability Bounds

- Method used to process a file fails w/ probability $\leq 1/2$.
- Make $17 \log_2 N = O(\log N)$ independent attempts to process one file:
 $\Pr[\text{failure on ALL attempts}] \leq (1/2)^{17 \log_2 N} = 1 / N^{17}$.
- We therefore succeed on this file with high probability.

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With failure probability of the form $1 / N^{\text{constant}}$
(the constant usually disappears in an $O()$ bound,
so it can be arbitrary)

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- We therefore succeed on this file with high probability.
- Now suppose you have N files to process.
- Run the approach above on all N files ($O(N \log N)$ total attempts)
- $\Pr[\text{failure on ANY file}] \leq N * \Pr[\text{failure on ONE file}] \leq 1 / N^{16}$.
- Still succeeds with high probability!

High Probability Bounds

What's a good choice for the type of bound we should adopt for the meaning of “with high probability”?

- $\Pr[\text{failure}] = \frac{1}{2}$
- $\Pr[\text{failure}] = 0.001$
- $\Pr[\text{failure}] = 1 / \log N$

-
- $\Pr[\text{failure}] = 1 / N^{17}$
 - $\Pr[\text{failure}] = 2^{-N}$



whp

Recall: The Randomized Reduction Lemma

- Suppose we have an algorithm for which:
 - We start with a problem of size n .
 - In every iteration, effective problem size is reduced to a constant fraction of its current value with some constant probability.
- Then, our algorithm performs only $O(\log n)$ iterations in expectation.
- **Example usage:** showing randomized quicksort spends $O(\log n)$ expected time on a generic input element (equivalent to randomized binary search).

The RR Lemma Also Gives High Probability Results!

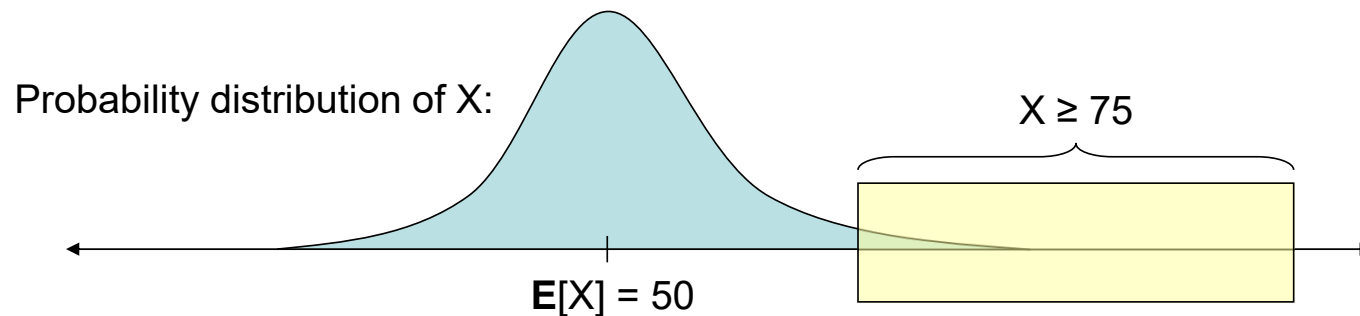
- Suppose we have an algorithm for which:
 - We start with a problem of size n .
 - In every iteration, effective problem size is reduced to a constant fraction of its current value with some constant probability.
- Then, our algorithm performs only $O(\log n)$ iterations in expectation and also with high probability!
- **Example usage:** showing randomized quicksort spends $O(\log n)$ time with high probability on a generic element.
- **Example usage:** showing randomized binary search runs in $O(\log n)$ time with high probability.

Randomized Quicksort Revisited: Per-Element Analysis

- Let's think again about randomized quicksort.
- We now have two ways to analyze its running time, each using slightly different tools...
- **Expected running time analysis:**
 - Randomized reduction lemma $\rightarrow O(\log n)$ expected time / element.
 - Linearity of expectation $\rightarrow O(n \log n)$ total expected time.
- **“With high probability” analysis:**
 - Randomized reduction lemma $\rightarrow O(\log n)$ time / element w.h.p.
 - Union bound $\rightarrow O(n \log n)$ time for all elements w.h.p.

Tail Bounds

- We're often interested in the probability that a random variable X deviates significantly from its mean, $\mathbf{E}[X]$, when we randomly sample from its distribution.
- Examples:
 - Show that $\Pr[T = O(\log n)] \geq 1 - 1/n^c$ for arbitrary constant $c \geq 1$.
 - What is $\Pr[X \geq 75]$ if X denotes the number of heads in 100 flips of a fair coin.



Markov's Inequality

- X is any *nonnegative* random variable, then
$$\Pr[X \geq k\mathbf{E}[X]] \leq 1/k.$$
- Sometimes written as $\Pr[X \geq a] \leq \mathbf{E}[X] / a.$
- Example: let X be the number of heads when flipping 100 coins.
 - $\mathbf{E}[X] = 50.$
 - $\Pr[X \geq 75] = \Pr[X \geq (3/2)\mathbf{E}[X]] \leq 2/3.$
- Example: If expected running time = T , then probability our algorithm takes $\geq 5T$ time is at most $1/5.$
- Due to its generality, Markov's inequality is a rather weak bound, although it's still quite useful.

Example

- Suppose you have an algorithm that runs in $O(n^2)$ worst-case time and takes $O(n)$ expected memory (say, $4n$ expected cells of memory).
- Memory is limited, so you want instead a worst-case bound on memory usage.
- Terminate and re-start algorithm if notice it using $>8n$ memory.
- $\Pr[\text{failure}] \leq \frac{1}{2}$, so $\Pr[\text{success}] \geq \frac{1}{2}$.
- $\mathbf{E}[\# \text{ runs}] \leq 2$, so running time now $O(n^2)$ in expectation.
(could also get $O(n^2 \log n)$ with high probability...)

Randomized Reduction – in Expectation

- Suppose we have an algorithm for which:
 - We start with a problem of size n .
 - In every iteration, effective problem size is reduced to a constant fraction of its current value in expectation.
e.g., $\mathbf{E}[\text{new problem size}] \leq 0.6$ (original size)
- Then, our algorithm satisfies the conditions of the randomized reduction lemma, so it performs only $O(\log n)$ iterations in expectation and with high probability.
 $\mathbf{Pr}[\text{new problem size} > 0.8 \text{ (original size)}]$
 $= \mathbf{Pr}[\text{new problem size} > (4/3) \mathbf{E}[\text{new size}]] \leq 3/4.$

Chernoff Bounds

- Suppose $X = X_1 + X_2 + \dots + X_n$ is a sum of independent indicator (0/1) random variables (a common scenario).
- Then the tails of X 's distribution drop off very quickly:
 - $\Pr[X \geq \mathbf{E}[X] + t] \leq e^{-2t^2/n}$
 - $\Pr[X \leq \mathbf{E}[X] - t] \leq e^{-2t^2/n}$
 - $\Pr[X \leq (1 - \varepsilon)\mathbf{E}[X]] \leq e^{-\varepsilon^2\mathbf{E}[X]/2}$
 - $\Pr[X \geq (1 + \varepsilon)\mathbf{E}[X]] \leq e^{-\varepsilon^2\mathbf{E}[X]/4}$ (if $\varepsilon \leq 2e - 1$)
 - $\Pr[X \geq (1 + \varepsilon)\mathbf{E}[X]] \leq 2^{-(\varepsilon+1)\mathbf{E}[X]}$ (if $\varepsilon > 2e - 1$)
- Example: If X denotes the number of heads we see when flipping 100 fair coins, $\Pr[X \geq 75] = \Pr[X \geq \mathbf{E}[X] + 25] \leq e^{-25^2/50} = e^{-12.5} \leq 0.000004$

Proving the “High Probability” Randomized Reduction Lemma

- **Theorem:** Suppose every iteration of our algorithm has probability $\geq p$ of reducing our problem size to at most $\leq q$ times its original size. Then given any constant $c > 0$, we can find another constant k such that $\Pr[T \geq k \log n] \leq 1 / n^c$, where T is the total number of iterations of our algorithm.
- **Proof:**
 - Suppose we run our algorithm for $k \log n$ iterations.
 - Let X denote the # of “good” iterations among these (a good iteration is one where we reduce the problem size to $\leq q$ times original).
 - Now note that $\Pr[T \geq k \log n] \leq \Pr[X \leq \log_{(1/q)} n]$, since the event that “ $T \geq k \log n$ ” is a subset of the event that “ $X \leq \log_{(1/q)} n$ ”.
 - So now let’s find a constant k such that $\Pr[X \leq \log_{(1/q)} n] \leq 1/n^c$.
 - Continued...